

# **Preparation Plan – Tech 3 (Master’s Level)**

**Instituto Itaú de Ciência, Tecnologia e Inovação**

**Inova Talentos**

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# Comprehensive Technical Structure

## Block 1 — Core Foundations

### Quantum Computing

- Quantum optimization (QAOA, VQE)
- Parameterized quantum circuits
- Simulation of quantum systems

### Mathematical Foundations

- Linear Algebra (vector spaces, operators, eigenvalues)
- Quantum Mechanics (postulates, Hermitian operators, measurement theory)

## Block 2 — Scientific Programming

- Advanced Python
- Scientific libraries:
  - NumPy
  - SciPy
  - Matplotlib
- Object-Oriented Programming (OOP)

## Block 3 — Quantum Frameworks

- Qiskit
- PennyLane
- Cirq

### Applications

- Quantum circuit simulation
- Execution on quantum hardware
- Integration with classical machine learning models

## Block 4 — Machine Learning and Generative AI

### Frameworks

- PyTorch
- TensorFlow
- JAX

### Models

- Transformers (LLMs)
- Variational Autoencoders (VAE)
- Diffusion Models

### Techniques

- Fine-tuning
- Model evaluation

## Block 5 — Optimization Methods

### Classical Optimization

- Gradient Descent
- Adam
- COBYLA
- SPSA

**Note:** Classical optimization plays a central role in hybrid quantum-classical algorithms, where parameters of quantum circuits are iteratively adjusted.

## **Block 6 — Quantum Machine Learning (QML)**

- Quantum Neural Networks (QNNs)
- Variational Quantum Eigensolver (VQE)
- Quantum Approximate Optimization Algorithm (QAOA)

### **Integrations**

- Qiskit Machine Learning
- PennyLane AI

## **Block 7 — Advanced and Desirable Skills**

### **Generative AI**

- Reinforcement Learning from Human Feedback (RLHF)
- Prompt engineering and dataset design

### **Infrastructure**

- GPU computing
- Experimental pipelines

### **MLOps**

- MLflow
- Weights & Biases
- Data Version Control (DVC)

## **Block 8 — Scientific Production and Reproducibility**

- Jupyter Notebooks
- LaTeX
- Git and GitHub
- Reproducible research practices

## **Learning Strategy (Realistic Roadmap)**

### **Stage 1 — Foundations**

- Scientific Python and linear algebra
- Fundamental quantum mechanics

### **Stage 2 — Intermediate Development**

- Quantum circuits using Qiskit or PennyLane
- Numerical optimization methods

### **Stage 3 — Advanced Topics**

- Variational algorithms (VQE, QAOA)
- Integration with machine learning frameworks (PyTorch, JAX)

### **Stage 4 — Specialization**

- Quantum Machine Learning (QML)
- Generative AI and end-to-end pipeline development



# Chapter 1

## Core Foundations

This block focuses on the fundamental structure of modern quantum computing:

**How to represent, evolve, and optimize quantum systems using linear algebra and parametrized quantum circuits**

### 1.1 QUANTUM COMPUTING (CORE)

Quantum computing is a computational paradigm that leverages the principles of quantum mechanics to process information. Unlike classical computers, which use bits as the basic unit of information, quantum computers use quantum bits (qubits), enabling fundamentally new forms of computation.

#### 1.1.1 Qubit and Hilbert Space

A qubit is a vector in a 2-dimensional complex vector space:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

with normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

This naturally introduces:

- Linear algebra

- Quantum probability

### 1.1.2 Operators and Quantum Gates

The evolution of a quantum state is given by:

$$|\psi'\rangle = U|\psi\rangle$$

where:

- $U$  is a unitary operator
- $U^\dagger U = I$

Examples of quantum gates:

- Pauli-X
- Hadamard
- CNOT

## 1.3 Quantum Measurement

Measurement probabilities:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2$$

After measurement, the state collapses to the observed eigenstate.

## 2. PARAMETRIZED QUANTUM CIRCUITS (PQC)

These are the foundation of:

- Variational algorithms (VQE, QAOA)
- Quantum machine learning



## General Form

$$|\psi(\theta)\rangle = U(\theta)|0\rangle$$

where  $\theta$  represents tunable parameters.

Typical parametrized gates:

$$R_x(\theta), \quad R_y(\theta), \quad R_z(\theta)$$

## Core Idea

Optimization problem:

$$\min_{\theta} \langle \psi(\theta) | H | \psi(\theta) \rangle$$

This is known as **variational optimization**.

## 3. VARIATIONAL QUANTUM EIGENSOLVER (VQE)

### Objective

To find the ground state energy of a Hamiltonian.

### Eigenvalue Problem

$$H|\psi\rangle = E|\psi\rangle$$

### Strategy

1. Choose a parametrized ansatz
2. Compute expectation value
3. Minimize using classical optimization

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

## Physical Interpretation

Approximation of the lowest eigenvalue of  $H$ .

## 4. QUANTUM APPROXIMATE OPTIMIZATION ALGORITHM (QAOA)

### Applications

- Combinatorial optimization (e.g., Max-Cut)

### State Construction

$$|\gamma, \beta\rangle = e^{-i\beta H_M} e^{-i\gamma H_C} |+\rangle$$

where:

- $H_C$ : cost Hamiltonian
- $H_M$ : mixing Hamiltonian

### Key Idea

Alternate between:

- Cost evolution
- Mixing evolution

This resembles:

- Trotterization
- Controlled quantum dynamics

## 5. SIMULATION OF QUANTUM SYSTEMS

### Time Evolution

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle$$

## Methods

- Exact matrix exponentiation
- Trotter-Suzuki decomposition
- Variational simulation

## Connection to Physics

Time evolution follows the Schrödinger equation:

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle$$

# 6. MATHEMATICAL FOUNDATION

## 6.1 Linear Algebra

### Complex Vector Spaces

- Orthonormal basis
- Inner product:

$$\langle \psi | \phi \rangle$$

### Linear Operators

Hermitian operators:

$$A = A^\dagger$$

These represent physical observables.

### Eigenvalues and Eigenvectors

$$A|v\rangle = \lambda|v\rangle$$

Applications:

- Energy levels
- Measurement outcomes

## Spectral Decomposition

$$A = \sum_i \lambda_i |v_i\rangle\langle v_i|$$

## 6.2 Quantum Mechanics

### Postulates

1. State: vector in Hilbert space
2. Observables: Hermitian operators
3. Measurement: eigenvalues
4. Evolution: unitary
5. Composite systems: tensor product

### Tensor Product

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

Foundation for:

- Entanglement
- Multi-qubit systems

### Measurement Operators

$$P_i = |v_i\rangle\langle v_i|$$

## FINAL CONNECTION

All concepts reduce to four pillars:

1. **Representation:** state vectors
2. **Transformation:** unitary operators
3. **Measurement:** Hermitian operators
4. **Optimization:** parameter tuning

## STUDY ROADMAP

### Phase 1 — Mathematics

- Complex vector spaces
- Inner product
- Eigenvalues and eigenvectors
- Hermitian operators

### Phase 2 — Quantum Mechanics

- Postulates
- Measurement theory
- Time evolution
- Composite systems

### Phase 3 — Quantum Computing

- Qubits
- Quantum gates
- Circuits

**Phase 4 — Variational Methods**

- Parametrized circuits
- VQE
- QAOA

**Phase 5 — Implementation**

- Python-based quantum frameworks
- Simulation and experimentation